

002 — Evidence Validation & Model Hardening

001's headline conclusions rest on a narrower base of `DEMONSTRATED` inputs than their stated precision implies — three payload denominators underpinning the sector's cost conclusions are `CLAIMED` rather than filed, the constitution's named binding constraint for orbital compute (F2) rests on an admitted placeholder and an unclosed heat-pump loop, and the platform's XBRL extraction carries a measured defect (DA-23) whose remedy has been applied to 12 issuer-quarters rather than the universe; validating these inputs will either confirm 001's conclusions within a quantified band or restate them, and the answer determines how much precision any downstream thesis is entitled to.

Constitution pin	1.5.0
As of	2026-09-18
Universe	SPCX 18% · RKLБ 16% · FLY 8% · BWXT 10% · GOOG 8% · VRT 8% · MSFT 6% · NVDA 6% · MRCY 5% · YSS 8% · UTHR 4% · SATS 5% · VOYG 4% · LUNR 3% · HAWK 3% · BA 3% · IRDM 4%
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The answer, up front

4 / 9 READINGS FIRE +3 land on the threshold with zero margin	46.47 PERCENTAGE POINTS one ratio, graded vs what its label names	14 / 17 ISSUERS STRIPPED a census of a default, not a defect	3 / 32 CORRECTIONS PROPAGATED recorded, correct, consumed by none
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The news: this thesis **falsified its own pillar**. PIL-5 asked whether at least half of 001's headline figures convert to **DEMONSTRATED**. Four of nine admissible readings fire; three more land exactly on the threshold with **zero margin**.

The reason is not that 001's numbers are wrong. **It is that a converted figure is not a converted basis**. A number can be reproduced exactly from filed cells and still not be the quantity its label names — off by **46.47 percentage points** on a single SPCX ratio.

What this means for the programme. DEMONSTRATED is not a sufficient grade for an input. Any thesis from 003 onward that cites this validated set must carry **the basis alongside the grade**, or it inherits a figure that reproduces and means something else. That is the widest-reaching result in this report.

What it does not mean. 001's arithmetic largely held. Per-ticker sums reproduced at IRDM, MRCY, UTHR and SPCX. What failed is **clearance and labelling** — a narrower and more fixable defect than "the model is wrong."

Method note. The spec never states the denominator, so the ledger publishes **every admissible reading** rather than selecting one. A falsifier that passes or fails on an unstated denominator is not falsifiable — publishing the ladder is the finding.

How to read this report. The falsification and its ladder come first, because they are the news; the defect register follows, then the corrections census, then the tooling caveats a reader needs before trusting any figure above, and finally the instructions for 003–011 and the gaps that remain. **The main line — "space is a cost curve, not a value pool" — is not overturned here; what this report moves is the evidentiary floor beneath it.**

Nine readings: four fire, three sit on the threshold

The falsifier is `share_of_001_headline_figures_converted_to_DEMONSTRATED < 0.5`. Six candidates for the denominator were identified; all nine resulting readings are reported.

Reading (denominator named)	Rows	Share	vs 0.5
All six headline lines	6	0.5000	not below — zero margin
same, CHK004-strict (Line 2 out)	6	0.3333	TRIGGERED
six plus the 12.3% claim	7	0.4286	TRIGGERED
the four value-bearing lines	4	0.7500	not below
same, CHK004-strict	4	0.5000	zero margin
every numeric claim in 001's register	27	[0.0000, 0.7778]	UNRESOLVED
SPCX — value / +components / +basis	12	0.8333 / 0.5000 / 0.2500	TRIGGERED
SPCX incl. null slot	13	0.7692 / 0.4615 / 0.2308	TRIGGERED
GOOG — cells / series	140 / 14	0.5643 / 0.1429	TRIGGERED

Four readings fire. Three land exactly on the threshold. Only two admit a comfortable pass — and both are the readings that count a figure as converted without asking what it is a figure of.

The register reading is inert by 78%. Twenty-one of twenty-seven of 001's numeric claims are already **DEMONSTRATED** and cannot move, so its uncertainty band is **wider than the threshold distance** — the reading cannot adjudicate the question it is being asked.

What this means. The pillar cannot be scored as passed by choosing a denominator: a falsifier that passes or fails on an unstated denominator is not falsifiable. **Any downstream thesis citing PIL-5 as satisfied is citing a reading, not a result** — which is why the ladder, not a single number, is the deliverable.

A converted figure is not a converted basis

One ratio, two answers, no basis field. SPCX operating_margin is served as +29.79%. It is a filed -16.68%. The difference is 46.47 percentage points — and both numbers are arithmetically correct.

What happened. The served value divides an AI-segment six-month loss by consolidated revenue. It is a real computation on real filed cells. It is also not the operating margin of the company, and the artifact that serves it names no basis — **which means the error is invisible to any check that recomputes the same way.**

Why this is the thesis's central finding. Every other defect in this report produces a number that is *wrong* — catchable by magnitude, by sign, by plausibility. This one produces a number that is **exactly right and answers a different question**. It survives every check the programme currently runs, because the check and the number are computed the same way.

Concept served by the platform	Bases the issuer actually reports
SPCX operating_margin	consolidated · AI segment · Space segment · Connectivity segment · year-ago segment
SPCX capex	compute-bearing class only · cash additions · total additions at cost · cash + Δpayable
GOOG diluted EPS	Class A · Class B · Class C · consolidated
UTHR gross margin	five bases, spanning 158.69 points, including two negatives

The instruction that follows. A downstream thesis must state the **basis**, not only the grade. Where a concept is reported on more than one basis and the platform collapses them with no basis field, the artifact cannot comply by diligence alone — **it must first discover that a second basis exists.**

The numbers held; the labels did not

001's per-ticker arithmetic was re-derived from filed cells in every case examined. It reproduced.

Issuer	What was checked	Result
IRDM	$225,237 - 191,229 = 34,008 \cdot$ $50,258 / 216,906 = 23.17\% \cdot +3.84\%$	correct
MRCY	revenue \$983.6M / \$912.0M / +7.9% · R&D \$59.7M / \$67.6M / -11.7%	reproduced exactly
UTHR	all ten per-ticker figures: -1.917% · -9.246% · +9.179% · +7.593% · +13.417% · 42.23%/45.64%	all reproduce
SPCX	\$15.0B reproduces exactly; the \$(1,257)M AI operating line is filed	confirmed

So the correction target is narrower than it looks. What failed is the *clearance* — the test that declared a figure clean — and the *labelling*. Not the sums. And the labels are what downstream theses price.

Two clearances failed a test that could not have failed. 001 cleared MSFT by testing only its **subtotal** facts, which are positive by construction for a profitable issuer — so the test passes regardless of what happened one level down, where two of three facts in a single row are in fact stripped. The same shape appears at IRDM, where 001's own clearance is refuted by four concepts confirmed stripped **in the very quarter it reads**.

The rule this produces. *A detector run only on the subject of a comparison cannot detect a comparison error.* **That means the repair target for 003 onward is the clearance step and the label — not the arithmetic, which largely reproduces.** MRCY is the proof: the component identity was run — on the period 001 was **reporting**, not the one it was **comparing against**. The reported period is one of the two filed-positive ones, so it passed by construction.

A census of a default

14 / 17 DA-23 EXHIBITED 2 subtotal-clean · 1 unevident · 1 unexercised	7 / 17 DA-29 / DA-30 ENGAGED 10 issuers never engage them	20 DA-26 COUNT moved 19 → 20, universality falsified
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DA-23 sign stripping is not a rare defect. A filed negative is served as a positive of identical magnitude. At RKLQ Q2 2026 the served OperatingIncomeLoss is +57,514,000 against a filed (57,514) thousand, where the served components give -57,514,000 — the filed cells are at [RKLQ 10-Q p.6](#). **Any screen on the raw field inverts: the worst loss-makers rank at the top.**

Entry	What it is	Incidence
DA-23	sign stripping, income statement <i>and</i> cash flow	14 / 17 — a default
DA-24	non-operating item inside the operating line	4 present · 11 refuted/clean
DA-25	normalised per-unit metrics	8 confirmed · 5 refuted · 2 not testable
DA-26	annual mislabelled as quarterly	9 confirmed · 1 falsifies universality
DA-27	fiscal labels from the calendar quarter	6 confirmed · 5 kind-4 · 4 refuted
DA-28	capital-structure discontinuity	8 present/confirmed · 2 not testable
DA-29	back-solved and opaque checks	7 engaged · 10 unengaged
DA-30	two bases, one concept, no basis field	7 engaged · 10 unengaged

The two entries added at 1.5.0 are engaged by 7 of 17 issuers and unengaged by 10 — and an unengaged cell is not a passed check. That is a coverage finding about the register, not about the issuers. **It means an unengaged cell must be scored as untested, never as clean** — otherwise the register's own coverage gap is read out as the platform's clean bill of health.

What this thesis changed about the register

Correction	Detail
The Clean row is subtotal-level only	MRCY, UTHR and SATS were removed from it. An issuer can be clean at every subtotal while a <i>component</i> of the same statement is sign-corrupted — BWXT is the first where both hold at once.
NVDA is UNEVIDENT, not clean	All 69 OperatingIncomeLoss facts are non-negative 2014–2026. The sign test can neither pass nor fail .
VRT is UNEXERCISED	61 of 61 facts positive; 13 of 13 identities close vacuously .
Stripping is not an income-statement phenomenon	VRT and NVDA are stripped on cash-flow subtotals . The discriminator is the sign of the value , not the concept and not the period.
DA-24's defining instance is inverted	SATS's 4.6× event is a non-cash impairment charge , not a sale — the licences remain on the balance sheet. Its independence proof is withdrawn .
DA-26's universality is falsified	The count moved 19 → 20 , and FLY is a counterexample .

Why the register needed correcting at all. The register's own census statistic tested *parent concepts*. A contaminant inside a filed subtotal is **absorbed by it** — so a census that brackets a subtotal **proves presence, never absence**. At SPCX the same measurement read "not evident" from a census that closed with zero residual, while the contamination sat inside the thing being measured. **That means every "clean" reading in the register is a statement about the census's reach, not about the issuer** — and a downstream thesis needs to know which one it is holding.

Three of thirty-two corrections propagated

The process finding. upstream_stale is mandatory in all 41 artifacts, **correct in all 41**, and **consumed by none**. A correction that is recorded and never propagated is indistinguishable in effect from one never made.

The measured case. 001 attributed a **72.0%** gross margin to UTHR. It is **AMGN's** number. UTHR's filed consolidated margin is **87.30%**, in a band of **82.93–89.24%** across nine periods. The barrier the microgravity case must clear is therefore **15.30 points higher** than stated — **the case is harder, not easier**.

That correction has now been derived three times and absorbed zero times. The upstream line still reads 72.0.

Correction	Propagated?
SATS DA-24 inversion — the founding instance is a charge, not a sale	yes — delivered to thesis 003
DA-23 census narrowed to subtotal-level	yes
VOYG sign strip reclassified	yes
UTHR 72.0% → 87.30%, a 15.30 pp barrier correction	no — derived 3×, absorbed 0×
001's falsifier-census membership (PIL-4, 5, 6 — not 3, 5, 6)	no
SPCX 12.3% — wrong on four independent axes	no

What this appears to require. A pin that records staleness and dispatches nothing is not a control — it is a comment. The remedy is a **resumption queue** discharged by name, not a field every artifact carries and no process reads.

Eleven falsifiers classified; the metric is zero

Every falsifier in 001 and 002 was classified against the disposition classes, **each with a named resolving source**. The pillar's own metric — `count_of_001_falsifiers_unclassified_or_without_named_resolving_source` — is **0**.

Class	Count	Character
UNRESOLVABLE-FROM-PUBLIC-SOURCES	9	no public source carries it — each names the disclosure that would resolve it
UNRESOLVABLE-FROM-PLATFORM	12	public and reachable in principle, not by this toolchain
REACHABLE-BUT-NOT-RECORDABLE	2	the datum is reachable; the contract cannot record it
UNVALIDATED-BY-PLATFORM	3	the validator ran and its result is not admissible
UNEXERCISED/UNEVIDENT/not-testable	31	each with a named kind and the kind's own remedy
testable, blocked by effort	1	on-platform work: ~19 issuer Item 1A reads

The distribution is the finding. The largest class is not "unresolvable" — it is **testable and not tested**: 31 cells a named remedy could move.

And two of the classes have different remedies that were being merged. "No public source carries it" needs monitoring. "Public but unreachable" needs platform reach. A third case exists where the datum is reachable and the **result cannot be recorded** — that needs a contract allowance, not a research task. **Reading them as one class means three different remedies get dispatched as one** — which is how 31 testable cells stayed untested.

The tooling reports success in cases where it did not run

These are platform behaviours, not issuer behaviours. Each one makes an **absence of signal** read as a pass.

Behaviour	Consequence
validate_calculation passes a wrong-signed value	Its reported column shares the stripped store , so it compares a stripped value to a stripped value. At SATS it returned pass on a figure the filing prints as negative.
It can also return zero rows	On a concept that is filed and seven arcs deep. A validator that never ran and one that ran and passed are the same row in every tally.
93% false-positive when it fails	Of 57 VRT rows, 39 fails yielded three true defects — and the true one is indistinguishable from the false ones by inspection.
computed is not a derivation	At NVDA that column carried a 37× error . It is not reproducible from the instrument's own returned tree.
reported is not the filed value	At NVDA it was a segment value; at MSFT a segment's 14,386 against a filed 155,237.
A us-gaap: zero is evidence about the tag	Filer extensions are silently missed — SATS's own impairment concept is one.

The pattern. In every case the failure mode is not a wrong answer — it is **no answer, reported as a clean one**. That is why this thesis's own tools now refuse to certify what they have not tested. **This means that for any downstream thesis a platform "pass" is a claim to be re-derived, not a clearance to be inherited.**

What downstream theses must change

The one instruction: carry the **basis**, not just the grade. A figure graded **DEMONSTRATED** may still not be the quantity its label names.

- 1. A grade is not a licence to quote.** Every input inherited from this validated set must name the basis it was computed on. Where the platform collapses two bases into one concept, the artifact must **discover the second basis first** — diligence alone cannot comply.
- 2. Recompute from components before using any operating figure.** DA-23 affects **14 of 17 issuers**. The remedy is the component identity shown in-line, and $\text{EPS} \times \text{shares}$ is **not admissible** as a sign test — it inverts in both directions.
- 3. Re-score 001's evidence count.** Three issuers fall out of the "clean" denominator on testability grounds alone, so 001's quoted count should be restated as *issuer-quarters where the test could actually run*.

Thesis	What this report changes for it
003	Received the SATS correction: 10.7% → 8.91%, and the event is an impairment charge , not a gain. Its cost-curve work rests on a figure whose direction was inverted.
004-010	Any segment or consolidated figure must carry its basis. The SPCX operating_margin finding applies directly to 004's SOTP.
011	Screens built on operating_income rank the worst loss-makers first unless recomputed. Break-even inputs are clean by construction; operating-line screens are not.

And the programme-level point. The main line — *"space is a cost curve, not a value pool"* — is not overturned here. What this thesis does is **narrow the evidentiary base it can be asserted from. This means the main line survives as a conclusion with a stated floor** — and 003-011 inherit that floor, not 001's four-significant-figure precision.

LIMITATIONS · WHAT COULD NOT BE RESOLVED

The three irreducible gaps

Gap	Why it cannot close	Class
SPCX facility-side draw	Not disclosed. A 216-page filing and 17 transcript pages carry zero hits for the relevant terms. The number will appear only when it is material — which is when it stops mattering.	UNRESOLVABLE-FROM-PUBLIC-SOURCES
FCC IBFS / ITU registries	Public and free , but outside this toolchain. The remedy is a connector , not monitoring — merging this class with "no source exists" would dispatch a monitoring task for an already-public source.	UNRESOLVABLE-FROM-PLATFORM
MRK, BMY, WWD operating income	The concept is absent or segment-only . No detector of any kind can run there, so the field is not wrong — it is unavailable , and absence must never be recorded as a passed check.	UNRESOLVABLE-FROM-PLATFORM

Three gaps, three different remedies. One is a monitoring task, one is a connector, one is a coverage hole in the data layer. Collapsing them into a single "insufficient data" label — which is what the programme did before this thesis — destroys the only information that makes them actionable. **This means the class label, not the missing number, is what the programme must carry forward.**

And one caveat on this report itself. Thesis 002's material is an **evidence audit**, not a sector thesis. Every finding here is a statement about **what can be reproduced from primary sources** — not about what the space economy is worth. The two questions are related but they are not the same question, and this report only answers the first.

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